Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs

Software-based detection methods — statistical tests, log-probability fingerprinting, benchmark comparisons — are fundamentally unreliable. Trusted Execution Environments offer the only provably secure path forward.

arxiv.org/abs/2504.04715

The Model Substitution Problem

Economic Incentives Drive Covert Swaps

1. Smaller Model
Serve 8B instead of advertised 70B — massive GPU cost reduction

2. Quantized Variant
FP8/INT8 instead of FP16 — near-identical benchmark scores

3. Fine-Tuned Version
Updated or distilled model sharing same family name

4. Different Family
Entirely different architecture serving same API endpoint

H_0: API serves the specified model → $P_{spec}(y|x)$
H_1: API serves a substitute → $P_{actual}(y|x) \neq P_{spec}(y|x)$

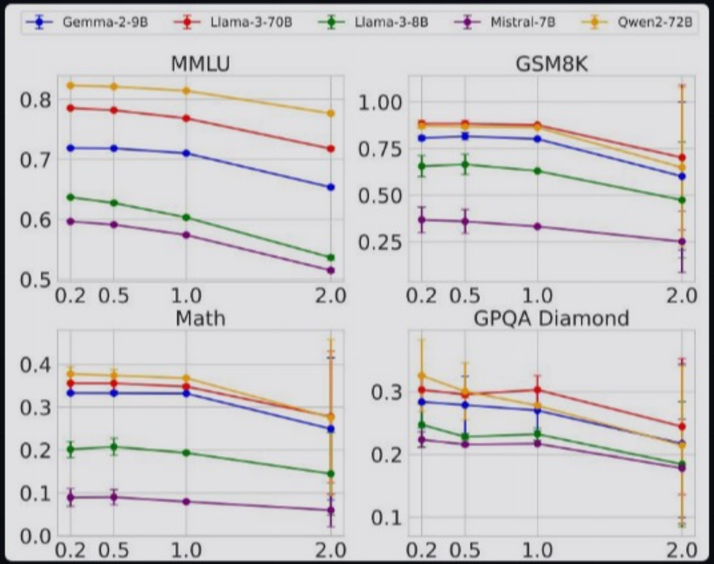
FP8 Quantization Preserves Most Accuracy

Making Substitution Hard to Detect

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
↳ FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
↳ FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
↳ FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
↳ FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
↳ FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Green = FP8 within original error bars | **Red** = statistically detectable gap — Most FP8 scores fall within error bars, especially on MMLU and GSM8K.

Higher Temperature Amplifies Gaps but Increases Noise



At **temperature 2.0**, all models degrade — the gap between original and substitute narrows or becomes noisy, complicating statistical detection across MMLU, GSM8K, MATH, and GPQA.

Log Probability Fingerprinting Fails

Inference Nondeterminism Defeats Metadata-Based Detection

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

The same **Llama-3-8B** model produces different log probabilities across software (transformers 4.40 vs 4.50, vLLM 0.6.2 vs 0.8.2) and hardware (A100 vs H100) — an auditor cannot distinguish substitution from infrastructure variation.

Randomized Routing at Low Rates

Statistical Detection Becomes Query-Prohibitive

$$P_{mixed}(y|x) = (1 - p) \cdot P_{spec}(y|x) + p \cdot P_{sub}(y|x)$$

As $p \to 0$, $P_{mixed} \to P_{spec}$ | Detection requires $O(1/p^2)$ queries

Specified Model $1 - p$ → API Response P_{mixed} ← Substitute p

Software-Only Auditing Is Fundamentally Unreliable

Method	Weakness	Practical Limitation
Text-Based Statistical Tests MMD	Require impractically large query budgets; fail against subtle substitutions where output distributions overlap significantly	Detection power drops to near-chance for FP8-quantized substitutes; query cost scales as $O(1/p^2)$ for randomized routing
Log-Probability Fingerprinting LOGPROB	Production inference nondeterminism across software versions and GPU hardware introduces uncontrolled variation	Cannot distinguish legitimate infrastructure variation from actual model substitution in real API deployments
Adversarial Countermeasures EVADE	Providers can detect benchmark prompts and route them to the genuine model; randomized mixing obscures signal	Any fixed audit protocol can be gamed; software-only approaches are vulnerable to adaptive adversaries

Each method fails independently — combining them does not recover reliable detection against an adaptive provider.

Trusted Execution Environments

Provable Cryptographic Model Integrity

🔒
Provable
Cryptographic attestation that correct weights are loaded and executed

⚡
Efficient
Modest performance overhead vs. prohibitive cost of ZKPs for large models

🛡
Robust
Resistant to all software-level evasion: randomized routing, benchmark evasion

Software Methods

- × Statistical noise masks subtle substitutions
- × Infrastructure variation breaks fingerprinting
- × Adaptive providers can game any fixed audit
- × No cryptographic guarantee of model integrity

Hardware Attestation (TEE)

- ✓ Verifiable proof of exact model weights
- ✓ Immune to software-level nondeterminism
- ✓ Cannot be gamed by adaptive adversaries
- ✓ Cryptographic attestation of inference code

Key Takeaways

Close the Verification Gap with Hardware Attestation

1. **Model substitution is economically motivated** and practically undetectable via software — providers save GPU costs with near-zero benchmark impact.
2. **FP8 quantization barely affects benchmarks** — most scores fall within original error bars, making it an attractive but nearly invisible substitution.
3. **Inference nondeterminism kills log-prob fingerprinting** — different frameworks and GPUs produce different outputs from the same model.
4. **Randomized substitution defeats statistical tests** — low-probability mixing requires exponentially more queries to detect.

The community must adopt hardware-attested inference to close the verification gap in commercial LLM APIs.